Saideep Katragadda

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Education

Texas A&M University **College Station, TX**

Bachelor of Science in Computer Science *Expected May 2028*

- **Relevant Coursework**: Python Computation, Business Information Systems Concepts, Discrete Structures, Program Design and Concepts, Data Structures and Algorithms, Principles of Statistics, Programming Languages

Technical Skills

Languages: Python, Java, C++, TypeScript, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, FastAPI, Tailwind CSS, PyTorch/TensorFlow, Scikit-Learn, Pandas
Tools & Infrastructure: Docker, PostgreSQL, Redis, Git/GitHub, Postman, Azure DevOps, VS Code, Eclipse
Certifications: IT Specialist - Java (Certiport), Microsoft Office - Word, Excel, PowerPoint, Access

Experience

Software Engineer Team Lead **October 2026 – Present**

Aggie Data Science Club (American Airlines Partnership) *College Station, Texas*

- Leading a **6-person team** to build a **time-series forecasting model** that predict airline market demand up to **2 years ahead** for **American Airlines Network Planning**, using GDD and demographic data.
- Building an **XGBoost regression baseline** on **300K+ flight records**, reaching $\mathbf{R^2 = 0.974}$ (MAE $\approx$ **2.1K**) on a 60K-record holdout set; feature-importance analysis shows **cabin class** drives $\sim$**91% of model gain**.

Undergraduate Researcher **October 2026 – Present**

Texas A&M University *College Station, Texas*

- Conducting research on an **agent-based model** (**Python**, **NetLogo**) of construction workers' exoskeleton adoption, parameterized from literature and expert input; using **sensitivity analysis** to identify drivers of **long-term adoption versus discontinuation** .

Application Developer Intern **June 2026 – August 2026**

University Federal Credit Union (UFCU) *Austin, Texas*

- Built AI-powered automation agents in **Azure DevOps** for sprint management and backlog tracking, plus a meeting intelligence pipeline that transcribed Microsoft Teams meetings and published AI-generated summaries to project wikis.
- Diagnosed and fixed failed lending requests in a banking loan **imaging API**, and built workflow automation with **Azure DevOps APIs** and **Azure cloud services** to cut project management work and improve documentation accuracy.

Quantitative Developer **March 2026 – Present**

Maroon Investment Club *College Station, Texas*

- Engineered a **dual-model XGBoost** classification system in **Python** to detect buy and sell opportunities during market dislocations, delivering a **36% backtested return** versus a **14% S&P 500** benchmark.
- Built a **feature-engineering pipeline** using **RSI**, **VIX**, moving averages, and momentum indicators, and applied **SMOTE** class balancing with **walk-forward validation** to reduce overfitting and improve out-of-sample performance.

Projects

CourtVision | *Python, Scikit-Learn, TypeScript, PostgreSQL, FastAPI, Next.js, Docker* **June 2026 - September 2026**

- Constructed a full-stack NBA analytics platform ingesting **131,000+ player game logs** across **6,151 games** and **26 draft classes**, with raw-cached, validated, and resumable data pipelines.
- Built a **random-forest prediction engine** and **10,000-run Monte Carlo simulation**, ranking actual MVP/ROY in the top 3 in **80% of backtested seasons**, with **94+% player-similarity** and **95% query-routing accuracy**.

Senna | *Python, PostgreSQL/TimescaleDB, Redis, Docker, Prometheus* **July 2026 - October 2026**

- Engineered a full-stack F1 telemetry analytics platform ingesting **19.8M+ samples** across **26,955 laps** into **PostgreSQL/TimescaleDB**, with $\pm$**35 ms lap alignment** and a **1,290+ test suite** at **90%+ coverage.**
- Designed a tyre-degradation forecaster (**0.37 s MAE**, leave-one-race-out validated) and a fact-validated AI engineer, serving cached comparisons at **89 ms p95** with **0 errors** across **26,649 load-test requests**.

Pixel Pathology | *Python, PyTorch/TensorFlow, Pandas, Streamlit, CNNs* **October 2026 - Present**

- Designing an interpretable CNN skin-lesion classifier in **PyTorch** on **10,015 HAM10000 images** (7 classes), fine-tuning **ResNet50/EfficientNet-B0** with two-stage transfer learning, and added **Grad-CAM heatmaps** to a **Streamlit demo** to show the regions driving each prediction.
- Building a leakage-safe data pipeline that groups **7,470 unique lesions** into seeded train/val/test splits and handles a **58:1 class imbalance** (67% nevi) with class-balanced sampling, scored on macro F1 and balanced accuracy.